Answer key - Summation and Series

GCE Advanced Level - Mathématiques - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. What does the sum from k = 1 to 4 of k equal?
Que vaut la somme de k = 1 a 4 de k ?

- 10 *
- 4
- 24
- 10 squared / 10 au carre

Answer: 10
1 + 2 + 3 + 4 = 10.

2. An arithmetic series has u1 = 5 and last term u(12) = 71. Find S(12).
Une serie arithmetique a u1 = 5 et pour dernier terme u(12) = 71. Trouve S(12).

Answer: 456
S(12) = 12 x (5 + 71) / 2 = 12 x 76 / 2 = 456.

3. Find the sum of the first 4 terms of the geometric series with u1 = 2 and q = 3.
Trouve la somme des 4 premiers termes de la serie geometrique de premier terme u1 = 2 et de raison q = 3.

Answer: 80
S(4) = 2 (1 - 3^4) / (1 - 3) = 2 (1 - 81) / (-2) = 80.

4. The arithmetic series has u1 = 2 and u(6) = 27. Which formula gives S(6)?
La serie arithmetique a u1 = 2 et u(6) = 27. Quelle formule donne S(6) ?

- S(6) = 6 x (2 + 27) / 2 = 87 *
- S(6) = 6 x (2 + 27) = 174
- S(6) = 2 + 27 = 29
- S(6) = 6 x 27 / 2 = 81

Answer: S(6) = 6 x (2 + 27) / 2 = 87
The arithmetic sum formula is n (u(1) + u(n)) / 2, so S(6) = 6 x (2 + 27) / 2 = 87.
La formule de somme arithmetique est n (u(1) + u(n)) / 2, donc S(6) = 6 x (2 + 27) / 2 = 87.

5. Find the sum of the first 5 terms of an arithmetic series with u1 = 4 and r = 6.
Trouve la somme des 5 premiers termes d une serie arithmetique de premier terme u1 = 4 et de raison r = 6.

Answer: 80
u(5) = 4 + 4 x 6 = 28, so S(5) = 5 x (4 + 28) / 2 = 5 x 32 / 2 = 80.
u(5) = 4 + 4 x 6 = 28, donc S(5) = 5 x (4 + 28) / 2 = 5 x 32 / 2 = 80.

6. Find the sum of the first 3 terms of a geometric series with u1 = 3 and q = 2.
Trouve la somme des 3 premiers termes d une serie geometrique de premier terme u1 = 3 et de raison q = 2.

Answer: 21

The terms are 3, 6, 12, so S(3) = 3 + 6 + 12 = 21. By formula: 3 (1 - 2^3) / (1 - 2) = 3 x (-7) / (-1) = 21.
Les termes sont 3, 6, 12, donc S(3) = 3 + 6 + 12 = 21. Par la formule : 3 (1 - 2^3) / (1 - 2) = 3 x (-7) / (-1) = 21.

7. What does the symbol Sigma with limits 1 and 4 express?
Que represente le symbole Sigma avec les bornes 1 et 4 ?

- The sum of the expression for the integers 1, 2, 3 and 4 / La somme de l expression pour les entiers 1, 2, 3 et 4 *
- The product of the expression for 1 and 4 / Le produit de l expression pour 1 et 4
- The difference between the fourth and first terms / La difference entre le quatrieme et le premier terme
- The average of the first four terms / La moyenne des quatre premiers termes

Answer: The sum of the expression for the integers 1, 2, 3 and 4
Sigma means a sum. The limits 1 to 4 mean you add the expression for each integer from 1 up to 4.
Sigma signifie une somme. Les bornes de 1 a 4 indiquent que l on ajoute l expression pour chaque entier de 1 a 4.

8. A geometric series has q = 1. Why does the sum formula need a special case?
Une serie geometrique a q = 1. Pourquoi la formule de somme a-t-elle besoin d un cas particulier ?

- Because the denominator 1 - q would be zero, and instead S(n) = n x u(1) / Parce que le denominateur 1 - q serait nul, et alors S(n) = n x u(1) *
- Because q must always be greater than 1 / Parce que q doit toujours etre superieur a 1
- Because the first term becomes zero / Parce que le premier terme devient nul
- Because the series stops being geometric / Parce que la serie cesse d etre geometrique

Answer: Because the denominator 1 - q would be zero, and instead S(n) = n x u(1)
When q = 1 the denominator 1 - q is zero, so the formula cannot be used. Instead every term equals u(1), so S(n) = n x u(1).
Lorsque q = 1 le denominateur 1 - q est nul : la formule est inutilisable. Dans ce cas chaque terme vaut u(1), donc S(n) = n x u(1).